

Question 6				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6				<u>Gene therapy techniques</u> A1 Insertion of a functional {gene/ DNA/ allele} (into cells)/ Replacement of a faulty {gene/ DNA/ allele} A2 Germ line {where (functional) gene added to embryo/ inherited} A3 Somatic gene therapy where functional gene is placed in body cells/ not inherited A4 Not permanent/needs re-treatment A5 Example of e.g.CF and CFTR gene inhaled into lungs/ Use of Restriction endonuclease/ligases/CRISPR/Cas9/stem cells/ Use of vectors/viruses/plasmids/liposomes <u>HGP link</u> B1 Aims of HGP: {sequence/ identify (position of)} {genes/ DNA} in the human genome/ improve understanding of genetic disorders/ improve the diagnosis and treatment of genetic disorders B2 {Prior to 2003/ before HGP sequenced} {very few/ fewer than 1,000} GT drug trials (per year). B3 {After 2003/ HGP sequenced} increase in GT trials B4 Reasons: (before 2003) Not all of human genome was sequenced/ (position of) {(faulty) genes/ mutations/ change in bases} was unknown/ ORA (for after 2003) B5 Up to 12,000 in 2017. <u>Use in gene therapy</u> C1 DMD is caused by a {mutation in the/ faulty} (dystrophin) gene and {failure to produce dystrophin/ protein too short/causes muscle wastage/weakness/ produces non-functional protein} C2 (Exon skipping introduces) a molecular patch over the exon with the {mutation/ fault} <i>Accept labelled diagram</i>	6	3		9		